ECE 376: Virtual Reality & Lab

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Spring 2026

Project #2 Written Proposal

Group Number: 09

Team members:

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Project Name:

- Baseball Batting Simulator

Professor: Yao Liu

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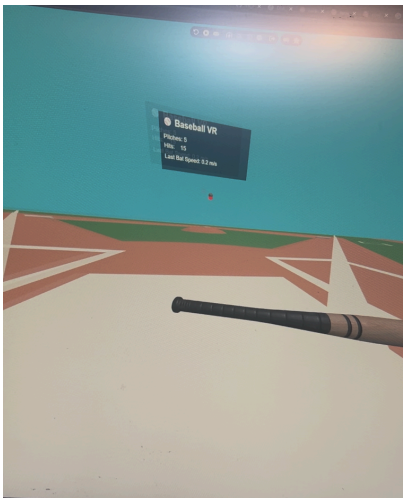
Introduction

Our project for this virtual reality experience is to make a fun and unique baseball batting simulator that is able to be played almost identically to how it would be in real life. What the goal is for this experience is to create a game where the user will be able to interact with a bat and spawn in a ball that can be swung at with the bat and hit. The user is able to hit as many balls as they want and see statistics on the most recently hit ball that show how far the ball went and the speed that it went. Virtual Reality makes this a better experience than a normal game due to the immersion we hope to add so that the player really feels like they are playing baseball rather than just hitting a ball in a blank space. Creating an environment that scales properly can make it so the user can be in a stadium hitting a baseball can make it a much more enjoyable experience and be easier to access than it would be in person due to cost, accessibility, and safety.

Environment Design



Our goal for the environment is to create an immersive baseball stadium so that the user feels like they are actually playing baseball in a meaningful place. This can be achieved through assets that are available along with creating a visual stadium that we plan on having generated based on a 360 image that we create. Currently we have a design of how the flooring would look that the user would be on that shows a baseball field as a prototype and once we continue working it will look like a realistic stadium as much as possible.

Interactions and Features

Our plan is to have the user use VR controllers to interact with and pick up a baseball bat. Once picked up, they can press a button that will allow them to spawn in a baseball that will be thrown from a distance where a pitching machine is and swing the bat they are holding to hit the ball once it is right in front of them. Once a ball is hit, they will see a scoreboard update with how many balls have been hit, how many balls have been spawned, and the distance / speed of the most recently hit ball.

Base Requirements:	
Theme	Creating a game that is a VR baseball batting simulator, allowing the user to spawn in and hit baseballs that are moving towards them.
Custom 3D Assets	- Baseball bat for the user to interact with
	- Pitching machine off in distance that the baseballs spawn from
Core Interactions	- Spawning in baseball via interaction with a button
	- Picking up a baseball bat and holding onto it to hit baseball that are thrown
Advanced:	
Category A - **Physics**	- Having collision that allows for a baseball to be hit realistically and change directions with Cannon-ES
Category B - **360-Degree Environment**	- Creating a stadium / sky using a 3d image to generate a background for the VR experience, adding immersion

Task Division

We will both evenly work on all aspects of this project, trying to split all work 50/50. Though it might be difficult to collaborate when doing this, we see benefit in working this way because it will allow us to understand what is going on in all stages of the process of building this VR simulation and help out one another easier than if we were to assign specific tasks. Many of the things that have two aspects like blender design and core interactions we will do one each and for the bigger things like actual creation of the game we will just work on everything together as much as possible.

Progress So Far

Currently, our progress with the game is making the initial layout of how we want everything to be: having baseballs spawn in with the press of a button and being able to swing an object to hit that ball. Some of our tracker details have also been completed which is all of the aspects that lay out the foundation of our game. Moving forward, we have to create the blender models we wish to use, create the 360 degree image for the setting, and tweak some of the physics to make it more realistic. Having this foundation makes it very easy for us to continue and finish this project.